

Tianyi Ouyang

Postdoctoral Research Associate, Carnegie Mellon University

✉ tianyio@andrew.cmu.edu ✉ touya001@ucr.edu ☎ +1 951-236-9409 🌐 [Homepage](#) 🎓 [Google Scholar](#)

Research Interests

Strongly correlated electronic and excitonic states in transition metal dichalcogenides (TMD) moiré heterostructures; Exciton polarons and exciton-based sensing of correlated phases; Magneto-optical spectroscopy; Interlayer excitons; Electrically tunable many-body phenomena in van der Waals heterostructures.

Research Experience

Carnegie Mellon University

Postdoctoral Research Associate, Department of Physics

Sep. 2025 – Present

Supervisor: Prof. Su-Fei Shi

University of California, Riverside

Ph.D. Researcher in Physics

Sep. 2019 – Jun. 2025

Advisor: Prof. Chun Hung Lui

Selected Publications

1. **T. Ouyang**, E. Liu, S. Cha, R. K. Paudel, Y. Sun, Z. Xu, T. Taniguchi, K. Watanabe, N. M. Gabor, Y.-C. Chang, and C. H. Lui, “Brightening Interlayer Excitons by Electric-Field-Driven Hole Transfer in Bilayer WSe₂,” *Physical Review Letters* **136**, 096903 (2026). (**Editor’s Suggestion**).
2. **T. Ouyang**, S. Cha, Y. Sun, T. Taniguchi, K. Watanabe, N. M. Gabor, and C. H. Lui, “Electrically Switching Ferroelectric Order in 3R-MoS₂ Layers,” *Nano Letters* **25**, 1459–1465 (2025).
3. **T. Ouyang**, Y. Meng, L. Yan, Y. Chen, S. Zhang, X. Chen, M. Erdi, T. Taniguchi, K. Watanabe, S. A. Tongay, B. Hunt, M. Xie, Y.-T. Cui, and S.-F. Shi, “Layer-Hybridized Wigner Crystals in MoSe₂/WS₂ Moiré Superlattice,” *arXiv:2608.01553* (2026). (*Nature Nanotechnology* under review)
4. S. Cha[†], Z. Xu[†], **T. Ouyang**[†], H. Yao, R. K. Paudel, T. Taniguchi, K. Watanabe, A. Y. Joe, Y.-C. Chang, N. M. Gabor, and C. H. Lui, “Hybridization-Revealed Interlayer 2p Exciton State in Bilayer MoS₂,” *Nature Physics*, accepted (2026).
5. **T. Ouyang** and S.-F. Shi, “Quantum Trions in Equilibrium,” *Science* **390**, 241 (2025).
6. S. Cha[†], **T. Ouyang**[†], T. Taniguchi, K. Watanabe, N. M. Gabor, and C. H. Lui, “Enhancing Resonant Second Harmonic Generation in Bilayer WSe₂ by Layer-Dependent Exciton-Polaron Effect,” *Nano Letters* **24**, 14847–14853 (2024).
7. S. Cha[†], **T. Ouyang**[†], T. Taniguchi, K. Watanabe, N. M. Gabor, and C. H. Lui, “Enhancing Resonant Second Harmonic Generation in Bilayer WSe₂ by Layer-Dependent Exciton-Polaron Effect,” *Nano Letters* **24**, 14847–14853 (2024).
8. E. Liu, M. Wilson, J. Hu, A. Zimmerman, A. Mathew, **T. Ouyang**, A. Shi, T. Taniguchi, K. Watanabe, T. F. Heinz, Y.-C. Chang, and C. H. Lui, “Exciton-polaron Umklapp scattering in Wigner crystals,” *arXiv:2601.11914* (2026).

[†] Equal contribution.

Technical Expertise

Cryogenic spectroscopy	Photoluminescence and reflectance contrast spectroscopy; magneto-optical measurements; experience with Montana and AttoDry cryostats.
Device fabrication	Dry transfer of van der Waals heterostructures; Electron-beam lithography; Electron-beam evaporation; Atomic force microscopy; Wire bonding.
Device control	Dual-gated TMD heterostructures with carrier-density and displacement-field tuning independently.
Data analysis & modeling	Python, Origin; Optical spectral analysis and fitting via physical models and numerical calculations of excitonic systems.

Education

Ph.D. in Physics, University of California, Riverside	2019 – 2025
Advisor: Prof. Chun Hung Lui	
B.S. in Physics, University of Science and Technology of China	2015 – 2019
Concentration: Theoretical Physics	Advisor: Prof. Zhenyu Zhang

Awards & Selected Presentations

Dissertation Completion Fellowship	2025
University of California, Riverside	
Outstanding Teaching Assistant Award	2024–2025
University of California, Riverside	
APS March Meeting — Presenter	2025
<i>Enhancing resonant second harmonic generation in bilayer WSe₂ by layer-dependent exciton-polaron effect</i>	
APS March Meeting — Presenter	2024
<i>Electric-field-switchable exciton polarons involving Γ-valley and K-valley Fermi sea in trilayer WSe₂</i>	
USTC Outstanding Student Scholarship, Silver Award	2016

Teaching Experience

Teaching Assistant, University of California, Riverside	Sep. 2019 – Dec. 2024
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References

Prof. Chun Hung Lui University of California, Riverside Ph.D. Advisor joshua.lui@ucr.edu	Prof. Yia-Chung Chang Research Center for Applied Sciences, Academia Sinica Research Collaborator yiachang@alum.sinica.edu.tw
Prof. Soonyoung Cha Sookmyung Women's University Research Collaborator scha@sookmyung.ac.kr	Prof. Nathaniel M. Gabor University of California, Riverside Ph.D. Committee / Research Collaborator nathaniel.m.gabor@gmail.com